

03_Data Cleaning & Preprocessing

Objective

The objective of this phase was to prepare the dataset for analytical use while preserving the integrity of real-world business events.

Rather than applying generic cleaning techniques, each data quality issue was investigated and evaluated within its business context to ensure that valid information was not unintentionally removed or altered.

Data Cleaning Strategy

The cleaning process focused on:

- Datatype validation and correction
- Missing value investigation
- Duplicate assessment
- Business-rule validation
- Data preservation

The primary goal was to improve analytical readiness while maintaining the original business meaning of the data.

Datatype Conversion

Several date-related fields were initially stored as text and required conversion to proper datetime format for accurate time-based analysis.

Datetime Conversions Performed

Table	Columns Converted
Orders	5
Order Items	1
Reviews	2

Total Conversions

8 Datetime Columns Converted

These conversions enabled accurate calculations for delivery timelines, fulfillment performance, order lifecycle analysis, and future feature engineering activities.

Missing Value Investigation

Missing values were analyzed individually to determine whether they represented data quality issues or legitimate business scenarios.

Orders Table

Null values were identified in delivery-related timestamp fields.

Investigation showed that these records were primarily associated with:

- Cancelled orders
- Unavailable orders
- Orders that never entered the fulfillment process

Decision

Retain Null Values

These missing values represent valid business events rather than data quality errors.

Reviews Table

Null values were identified within review comment fields.

Investigation revealed that many customers submitted ratings without providing written feedback.

Decision

Retain Null Values

The missing values reflect normal customer behavior and do not indicate incomplete records.

Products Table

Null values were identified within product metadata fields.

Investigation confirmed that affected products were associated with legitimate transactions and sales activity.

Decision

Retain Null Values

Removing these records would result in the loss of valid business transactions.

Duplicate Assessment

Duplicate checks were performed throughout the dataset.

No critical duplicate key issues were identified within the primary business entities.

The dataset maintained structural consistency and did not require record removal.

Cleaning Outcome

Final Results

Rows Dropped: 0
Values Imputed: 0
Datetime Columns Converted: 8

No records were removed during the cleaning phase because all investigated anomalies were determined to have valid business explanations.

Key Insight

The cleaning process demonstrated that effective data preparation is not simply about removing null values or deleting records. Each issue must be evaluated within its business context before corrective actions are applied.

By preserving legitimate business events and correcting only structural issues, the dataset remained both accurate and analytically reliable.

Outcome

The dataset was successfully prepared for feature engineering and exploratory analysis while maintaining complete transactional integrity.

The resulting tables were approved for advanced analytics, SQL integration, KPI development, and dashboard reporting.